

Mekdela Amba University
College of Business and Economics
Department of Economics

Model exam for 4th year Economics students

Name: D40 ✓

ID No: 12

Section: A

Max. no of questions: 60

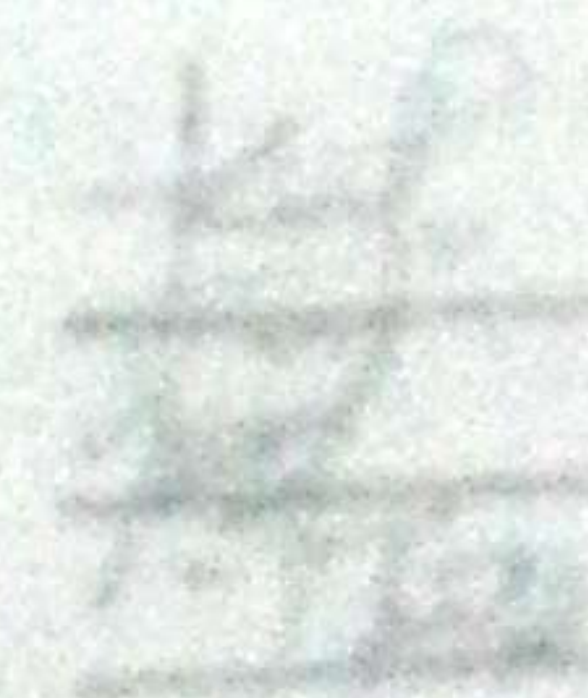
Time Allowed: 2.30

Instructions:

- Attempt all questions
- Cheating is strictly forbidden
- Use your time effectively

Approved by the committee-

1. Kunneger T.
2. Wondimu B.
3. T/Silasie K.





1. DEVELOPMENT ECONOMICS-I

- 1) The idea that poor countries themselves are becoming increasingly aware of their own backwardness, which led to a natural desire to for more rapid economic growth and greater political recognition through greater economic strengthen goes with
 - A) Academic interest for development economics
 - B) Mutual interdependency of world economy
 - C) Mutual dependency of world economy
 - D) New economic order
- 2) All are correct according to the Harrod Domar growth model **except**
 - A) does not explain why growth rates at different level of per capita income differ
 - B) Capital output ratio is assumed to be fixed
 - C) Saving and investment are a sufficient condition for growth
 - D) fundamental of growth (saving, capital output ratio, and population growth rate) are endogenous variable
- 3) Capital output ratio, θ , in the Harod Domar model is
 - A) a measure of productivity of capital
 - B) the amount of capital required to produce a single unit of output in the economy
 - C) the ratio of $Y(t)$ to $K(t)$ at given period of time t
 - D) the rate at which capital depreciates
- 4) The rate of saving in Solow model
 - A) does not affect the long-run growth
 - B) doesn't affects the long-run level of income
 - C) does not affect the short-run growth
 - D) does not affect the short-run level of income
- 5) In the solow model of unconditional convergence all are true **except** one
 - A) there is a negative relationship between initial per capita income and per capita income growth rate,
 - B) poorer countries grow faster
 - C) poor countries have lower marginal return to capital

- D) countries converge to the same steady state
- 6) The Harrod-Domar growth model suggests that growth is
- A) Indirectly related to savings and the capital/output ratio
 - B) Directly related to savings and the capital/output ratio
 - C) Directly related to savings and inversely related to the capital/output ratio
 - D) Directly related to the capital/output ratio and inversely related to savings
- 7) The Lewis model explains how growth gets started in a less developed economy
- A) With an average product of labor in agriculture that is negative
 - B) With a downward-sloping supply curve of labor
 - C) With a marginal productivity of labor zero or negligible in industry
 - D) With a traditional agricultural sector and an industrial capitalist sector
- 8) Migration of human capital and entrepreneurs from countries where reward and return for labor is lower to higher, which result in depression demand for goods and services in poorer country in the Process of Cumulative Causation is called
- A) backwash effect
 - B) spread effect
 - C) Spillover effect
 - D) Trickle-down effect
- 9) According to Simon Kuznets, the relationship between GNP per capita and inequality in the distribution of income can be expressed as
- A) A strictly decreasing relationship
 - B) A strictly increasing relationship
 - C) No relationship
 - D) First increasing and then decreasing
- 10) The vicious theory states that
- A) Growing government assistance create addiction to welfare programs
 - B) Low income levels create pressure for money creation
 - C) Low income levels create pressure for cheap imports
 - D) Low per capita income creates low saving that keeps incomes low

2. ECONOMETRICS-I

1. Which one of the following regression model shows the assumption of linear in parameters?
[A]. $\ln Y_i = \beta_0 + \ln \beta_1 X_i + U_i$
[B]. $Y_i = \beta_0^2 + \beta_1^2 X_i + U_i$
[C]. $\ln Y_i = \beta_0 + \beta_1 \ln X_i + U_i$
[D]. $Y_i = \ln \beta_0 + \beta_1 \ln X_i + U_i$
2. The residual is defined as the difference between the _____.
[A]. Actual value of y and the estimated value of y
[B]. Actual value of x and the estimated value of x
[C]. Actual value of y and the estimated value of x
[D]. Actual value of x and the estimated value of y
3. Which of the following statements is true about the simple linear regression model?
[A]. Linear relationship between a Regressand and two explanatory variables.
[B]. Linear relationship between a regressor and two dependent variables
[C]. Linear relationship between a Regressand and two or more explanatory variables
[D]. Linear relationship between explained and an explanatory variable
4. Which one of the following is **not true** about assumptions of simple Linear Regression Model?
[A]. $E(U_i) = 0$
[B]. $Cov(Y_i, X_i) = 0$
[C]. $Var(U_i) = \sigma^2$
[D]. $Cov(X_i, U_i) = 0$
5. Econometric model is different from mathematical model because:
[A]. Econometric model shows exact relationship
[B]. Econometric model shows deterministic relationship
[C]. Econometric model shows stochastic (inexact) relationship
[D]. None.
6. The aim of Econometrics is/are;
[A]. Formulation of economic models.
[B]. Specification of economic models.
[C]. Estimation and testing of econometric models
[D]. All.
7. The primary objective in regression analysis is:
[A]. To estimate population regression function
[B]. To estimate sample regression function
[C]. To develop theory
[D]. All.
8. The econometric model such that $Y = \beta_0 + \beta_1 X_1 + U_i$ where the disturbance term included unlike in mathematical model. The reason of introducing U_i can be because of...
except
[A]. Correct specification of the mathematical model
[B]. The erratic nature of human beings
[C]. Error of aggregation.
[D]. All.
9. Which of the following is **not true** about the decision of hypothesis testing?
[A]. If $t^* > t_c$, reject H_0 , $\hat{\beta}$ is statistically significant

- [B]. If $t^* < t_c$, accept H_0 , $\hat{\beta}$ is statistically significant
- [C]. $se(\hat{\beta}_i) < \frac{1}{2}(\hat{\beta}_i)$, reject H_0 , $\hat{\beta}$ statistically insignificant
- [D]. $se(\hat{\beta}_i) < \frac{1}{2}(\hat{\beta}_i)$, accept H_0 , $\hat{\beta}$ statistically insignificant
10. Which of the following desirable properties of an econometric model should be compatible with the postulates of economic theory.
- [A]. Explanatory ability
- [B]. Theoretical plausibility
- [C]. simplicity
- [D]. Forecasting ability

3. INTERNATIONAL ECONOMICS-I

- A primary reason why nations conduct international trade is because:
 - Some nations prefer to produce one thing while others produce another
 - Resources are not equally distributed to all trading nations
 - Trade enhances opportunities to accumulate profits
 - Interest rates are not identical in all trading nations
- One of the following is not included in the study of international trade
 - Supply- demand analysis
 - Consumer behavior
 - Exchange rate
 - Market structure
- Which one of the following is not the importance of international economics
 - It helps to identify global economic issues
 - It helps to understand the functions and objectives of international institutions
 - It helps to understand world economic interdependence
 - None of the above
- If a country has a linear (downward sloping) production possibilities frontier, then production is said to be subject to:
 - Constant opportunity costs.
 - Decreasing opportunity costs.
 - First increasing and then decreasing opportunity costs.
 - Increasing opportunity costs
- Once certain industries are given protection, they claim it as a matter of right. This statement explains one of the following causes of arguments against trade protection
 - Protection produces lethargy and acts like an opiate
 - Vested interests are created
 - There is the danger of corruption
 - Protection creates monopolies

6. Which one of the following will happen when a nation moves from free trade to tariff situation?
- A. An increase in the producers' surplus
 - B. An increase in the consumers' surplus
 - C. Both consumers' and producers' surplus increase
 - D. A decrease in the government revenue
7. The terms of trade is given by the prices:
- A. Paid for all goods exported by the home country.
 - B. Received for all goods exported by the home country.
 - C. Received for exports and paid for imports.
 - D. Of primary products as opposed to manufactured products.
8. The opportunity cost of producing one more unit of product A is equal amount of producing the other product B no matter what level of product A the country is producing. This can explain:-
- A. Constant cost condition
 - B. Complementarity of factors of production
 - C. Differences in factor quality
 - D. All of the above
9. Which one of the following is not among the assumption of factor endowment theory?
- A. Demand conditions are the same among nations
 - B. Nations use factor inputs that are uniform quality
 - C. Two countries produce using different production functions
 - D. Nations use the same technology
10. Which one of the following is not an argument for protection?
- A. Infant industry argument
 - B. Diversification of industry argument
 - C. Revenue argument
 - D. Inefficiency in world resource allocation argument

4. MACROECONOMICS -I

1. In Quantity theory of money, if velocity of money is kept constant:
 - A. Money supply determines the nominal value of output
 - B. Money supply will have no effect on output
 - C. Supply of real money balance decreases
 - D. None of the above
2. One of the following is not a factor that can affect net export of a country;
 - A. Income of consumers at home and abroad
 - B. Taste of consumers for domestic and foreign goods
 - C. Exchange rate
 - D. Transportation cost
 - E. None of the above
3. Assuming Ethiopia is under fixed exchange rate system, if Birr buys more foreign currency, we can say that;
 - A. There is depreciation of birr
 - B. There is appreciation of birr
 - C. Value of Ethiopian currency is decreasing
 - D. There is revaluation of birr
 - E. A and C
4. One of the following is correct about IS-LM and Mundell Fleming models
 - A. Both models focus on the interaction between the commodity and money market
 - B. Both models assume small open economy with perfect capital mobility
 - C. Both models assume that the price level is endogenously fixed
 - D. B and C
 - E. All of the above
5. Which of the following is false about contractionary monetary policy;
 - A. Real interest rate increases
 - B. Planned investment falls
 - A. The monetary base declines
 - C. Aggregate demand increases
6. using government spending and taxation to for the purpose of stabilizing the economy is called;

A. budget policy

C. trade policy

B. monetary policy

D. fiscal policy

7. A curve that shows the relationship between unemployment and inflation is called;
- A. Aggregate demand curve
 - B. Consumption curve
 - C. Indifference curve
 - D. Phillips curve
8. On the Keynesian cross, when actual expenditure is greater than planned expenditure;
- A. Firms accumulate inventories and production increases
 - B. Firms accumulate inventories and decrease production
 - C. Firms reduce inventories and production increases
 - D. Firms reduce inventories and decreases production
9. One of the following is **correct** about money market in small open economy;
- A. demand for real balances depends positively on the interest rate
 - B. Money supply and price level are assumed to be endogenously fixed
 - C. Demand for real balance depends positively on income level Y
 - D. None of the above
10. Which of the following is **false** about contractionary monetary policy and its effect;
- A. Real interest rate increases
 - B. The monetary base declines
 - C. Planned investment falls
 - D. Aggregate demand increases

5. MATHEMATICAL ECONOMICS

1. The function $h(x, y, w) = 2x^2 + 3yw - w^2$ is

- A) Homogenous of degree 2; doubling of all variables will double the value of the function.
 - B) Homogenous of degree 2; doubling of all variables will quadruple the value of the function.
 - C) Homogenous of degree 1; doubling of all variables will double the value of the function
 - D) Homogenous of degree 2; doubling of all variables will halve the value of the function
2. The utility a consumer derives from consuming two goods A and B can be assumed to be determined by the utility function $U = 40A^{0.5}B^{0.5}$. If A costs birr 4 per unit and B costs birr 10 per

unit, given the consumer income equal to birr 600, the combination of A and B that maximize consumers utility is:

- A) 40 unit of good A and 44 unit of good B
- B) 50 unit of good A and 40 unit of good B
- C) 60 unit of good A and 36 unit of good B
- D) 45 unit of good A and 42 unit of good B

3. The general solution for the differential equation $\frac{dy}{y^2} = xdx$ is

- A) $y = \frac{-2}{x+c}$
- B) $Y = \frac{-x}{2+c}$
- C) $y = \frac{-c}{x+2}$
- D) $y = \frac{-x}{2c}$

4. The equation $3y_{t+1} + 5y_t = 10$ is

- A) Homogenous linear first order difference equation
- B) Autonomous linear first order difference equation
- C) Non-autonomous linear first order difference equation
- D) Autonomous non-linear first order difference equation.

5. A Taylor series of the function $f(x) = x^2 + 3x + 2$ at $x = 1$ can be written as:

- A) $5(X - 1) + (X - 1)^2$
- B) $6 + 5(X - 1) + (X - 1)^2$
- C) $2 - 5(X - 1) + (X - 1)^2$
- D) $2 + 5(X - 1) + (X - 1)^2$

6. A profit $P(q)$ from producing q unit of good is given by $p(q) = 396q - 22q^2 + K$ for some constant K . using linear approximation about $q=80$ we found $p(81) \approx 17244$. Then, the value of K is;

- A) 400
- B) -400
- C) 300
- D) -300

7. The differential equation $\frac{d^2y}{dx^2} + 5y\left(\frac{dy}{dx}\right)^3 - 2x = 0$ is classified as

- A) Second order; first degree; non-linear and homogenous ODF
- B) Second order; first degree; linear and homogenous ODF
- C) Second order; first degree; non-linear and non-homogenous ODF
- D) First order; third degree; non-linear and non-homogenous ODF

8. The total revenue in birr received from the sale of X unit of product is given by $R(X) = 3x^2 + 36x + 5$. The marginal revenue when $X=15$ is

- A) 116
- B) 90
- C) 96
- D) 126

9. If the function $f(x) = 2x^2 - kx + 5$ is increasing on the interval $[1, 2]$, then k lies in the interval:

- A) $(-\infty, 4)$
- B) $(4, \infty)$
- C) $(8, \infty)$
- D) $(-\infty, 8)$

10. If the first five terms of the Taylor expansion for $f(x)$ about $x = 0$ are $3 - 7x + \frac{5}{2}x^2 + \frac{3}{4}x^3 - 6x^4$, then $f'''(0) =$

- A) $1/8$
- B) $9/2$
- C) $3/4$
- D) 6

6. MICROECONOMICS-I

1. If a rational consumer is in equilibrium, then:
 - A. the marginal utility obtained from one product is equal to the marginal utility obtained from any other product.
 - B. a reallocation of income would increase the consumer's total utility.
 - C. the marginal utility per last dollar spent is the same for all goods consumed.
 - D. total utility becomes zero
2. Identify the true statement from the followings.
 - A. We can use risk and uncertainty interchangeably
 - B. Risk has many possible outcomes and the probability to each outcome is not known
 - C. certainty has many possible outcomes and the probability to each outcome is known
 - D. from two products, product with less variance value is chosen in deciding which one to choose
3. An isoquant curve illustrates
 - A. All the alternative combinations of two products among which a producer is indifferent because they yield the same profit
 - B. All the alternative combinations of two products that can be produced by using a given set of inputs fully and in the best possible way.
 - C. All the alternative combinations of two inputs that yield the same maximum total product.
 - D. All the alternative combinations of two output produced using same level of two inputs
4. The law of diminishing returns implies that, with the use of capital fixed, as the use of labor rises,
 - A. the marginal product of labor will fall eventually.
 - B. total product will fall eventually.
 - C. the production process will become technologically inefficient eventually.
 - D. the total product of labor will fall below the marginal product of labor.
5. What is the difference between perfect competition and monopolistic competition?
 - A. Perfect competition has a large number of small firms while monopolistic competition does not.
 - B. In perfect competition, firms produce identical goods, while in monopolistic competition, firms produce slightly different goods.
 - C. Perfect competition has no barriers to entry, while monopolistic competition does.

- D. Perfect competition has barriers to entry while monopolistic competition does not.
6. The underlying principle behind a Profit maximizing firm using only one variable input will produce in stage-II is that
- A. Marginal product of variable input is greater than average variable input.
 - B. Marginal product of variable input is positive and increasing.
 - C. Marginal product of variable input is decreasing while fixed input is utilized efficiently.
 - D. Variable input is efficiently utilized due to increasing marginal product of variable input.
7. The magnitude of the slope of an indifference curve is the
- A. marginal rate of substitution.
 - B. rate of increasing opportunity cost.
 - C. marginal rate of utility of income.
 - D. rate of relative prices.
 - E. marginal utility of substitution
8. A point where the budget line is just touching an indifference curve at one point is
- A. the best affordable point.
 - B. the least affordable point.
 - C. on the lowest attainable indifference curve.
 - D. Both answers B and C are correct.
 - E. Both answers A and C are correct.
9. Which of the following is an example of an implicit cost?
- A. the economic depreciation of capital equipment the business owns
 - B. the cost of fuel and materials.
 - C. wages paid to workers
 - D. rent on a building
 - E. the cost of fertilizer for a farmer
10. The long run is defined as
- A. the period of time when most (more than 50 percent) resources are variable.
 - B. any time after six months.
 - C. any time after one year.
 - D. the period of time when all resources are fixed.
 - E. the period of time when all resources are variable.

Answer Sheet

Development I

1. D
2. D
3. B
4. A
5. C
6. C
7. D
8. A
9. D
10. D

Econometrics
I

1. C
2. A
3. D
4. B
5. C
6. D
7. A
8. A
9. B
10. B

By: D4D
MAU

International
I

1. A
2. B
3. D
4. A
5. B
6. A
7. C
8. A
9. C
10. D

Macro I

1. A
2. E
3. D
4. A
5. C^R = D
6. D
7. D
8. B
9. C
10. D

Mathematical
Economics

1. B
2. B
3. B
4. B
5. B
6. B
7. A
8. D
9. A
10. D

Micro
I

1. A
2. A
3. C
4. A
5. B
6. D
7. A
8. A
9. A
10. E

D4D Economics